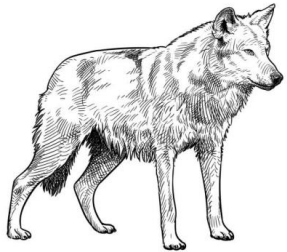
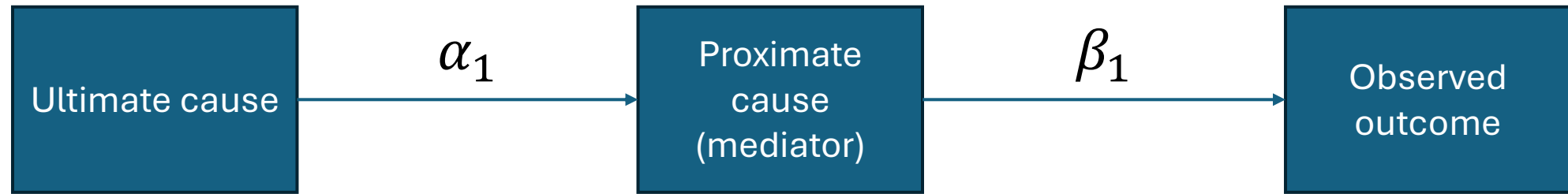


Path analysis: just more than one linear model!



Reminder: 24 September

Project timeline

Date	Assignment
24 Sep	Topic summary
1 & 3 Dec	Oral presentations
10 Dec	Paper due

Topic Summary

- 1-2 pages
- Describe your project (very briefly, i.e., study species, research questions, covariates) and outline why (or why not) you wish to use a structural equation model to analyze your data
- It may be (very) helpful to include a ‘causal diagram’

Unintended or unforeseen consequences



Malaria is spreading across Borneo



WHO sprays DDT



DDT kills mosquitos



Parasitic Wasp dies off



Caterpillars Flourish



Caterpillars eat thatch roofs



DDT consumed by flies



Geckos eat flies and pass DDT



Cats eat Geckos and die from DDT



Rats survive and spread the plague



Actions typically don't have a solitary consequence

- Second- and third- (and fourth- and fifth-) order effects
- Knock-on effects
- Systemic reverberations
- Externalities
- Side or ripple effects
- Whoopsie-doodles

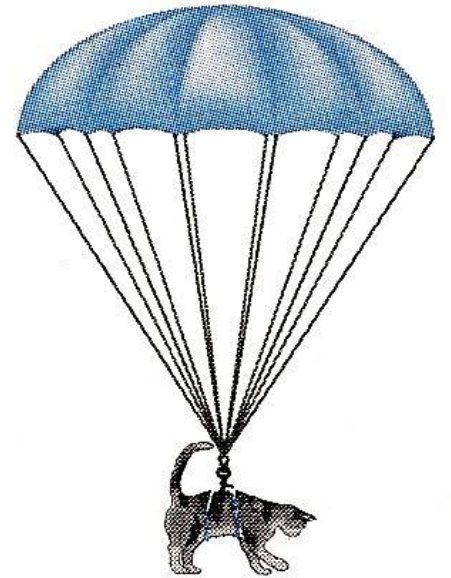
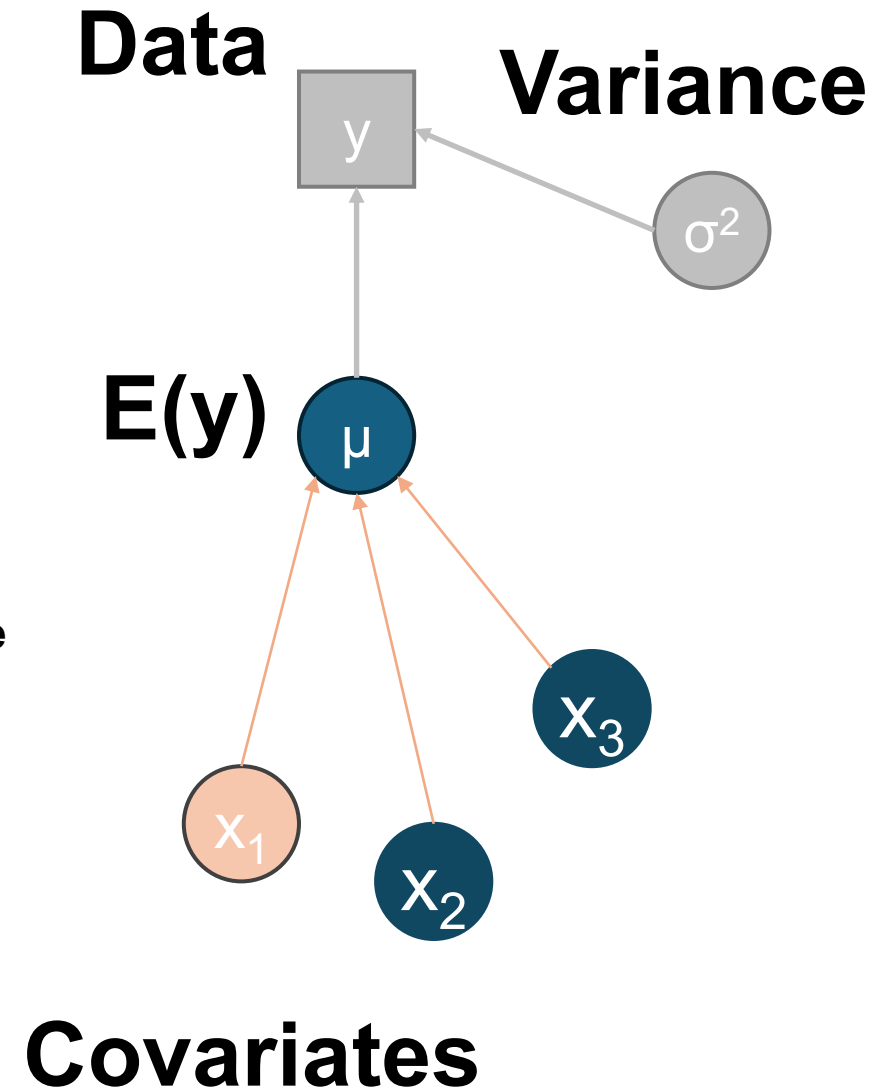
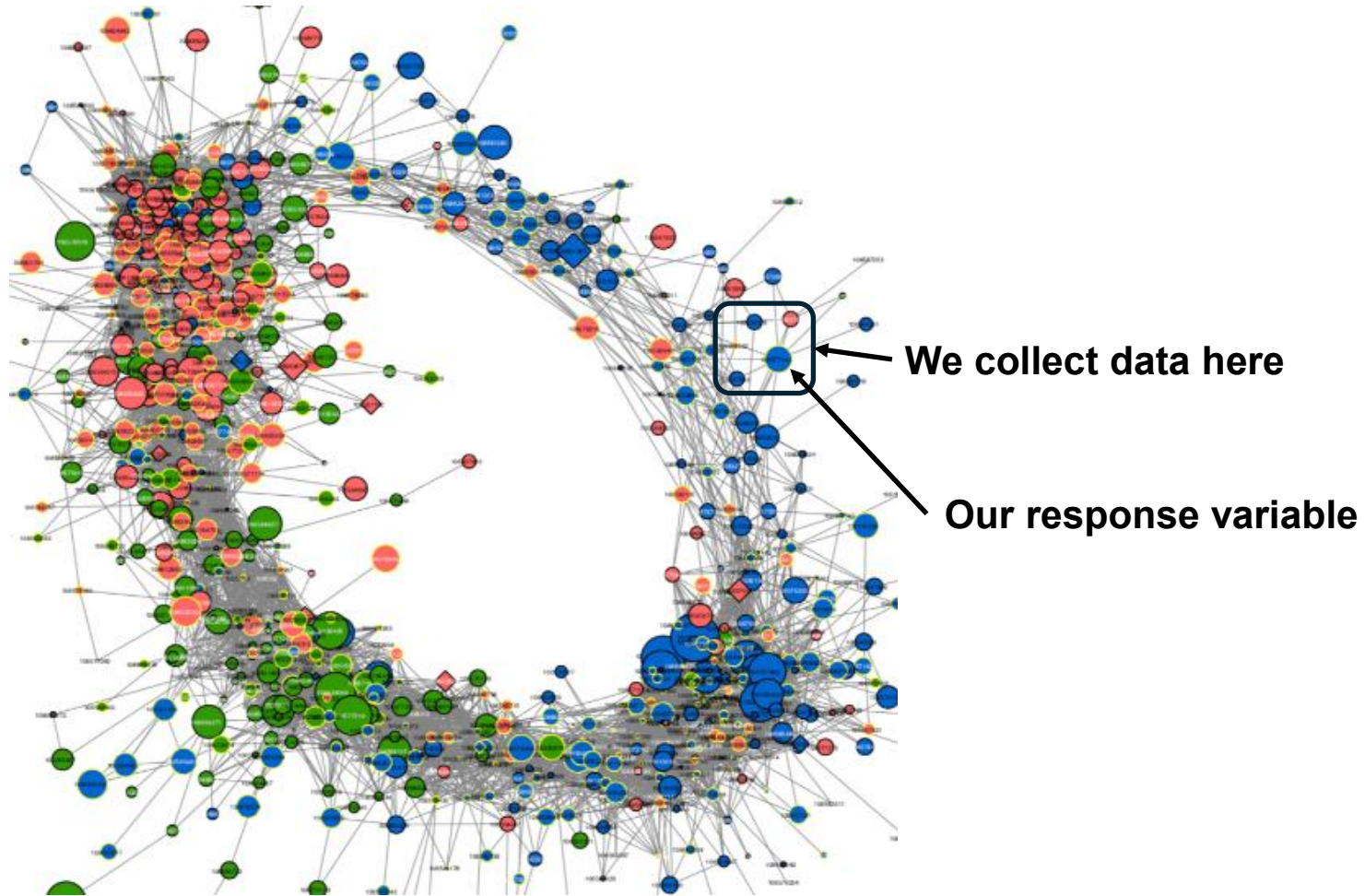


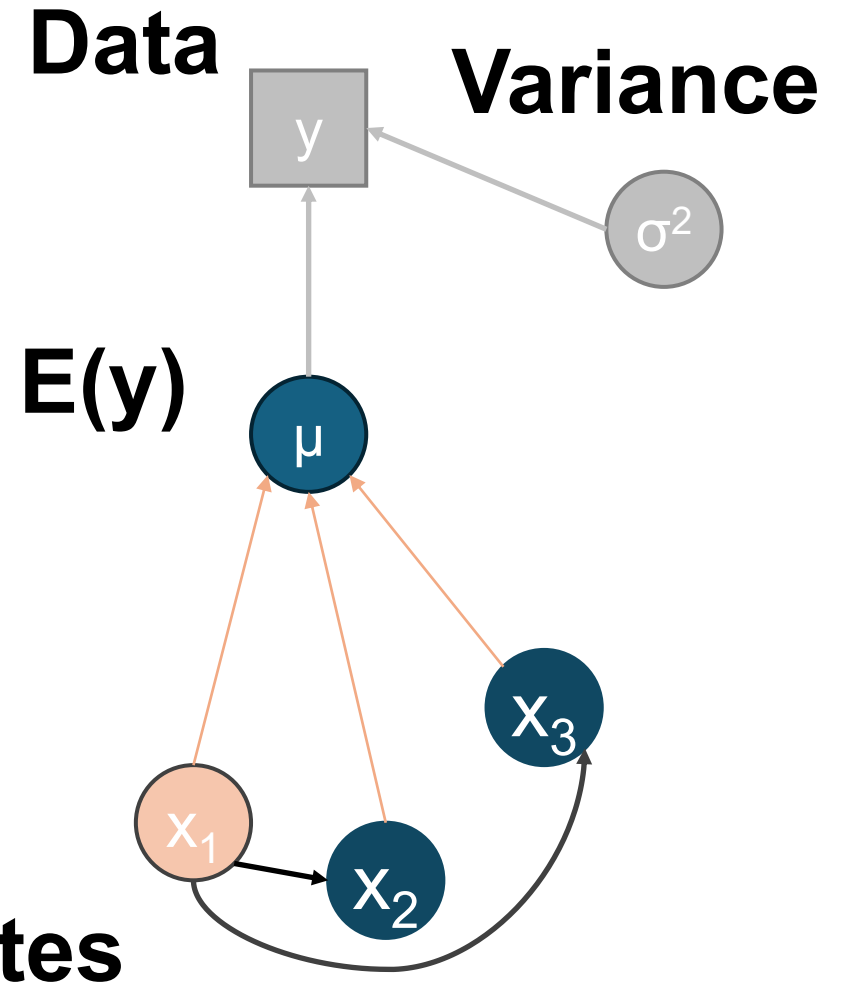
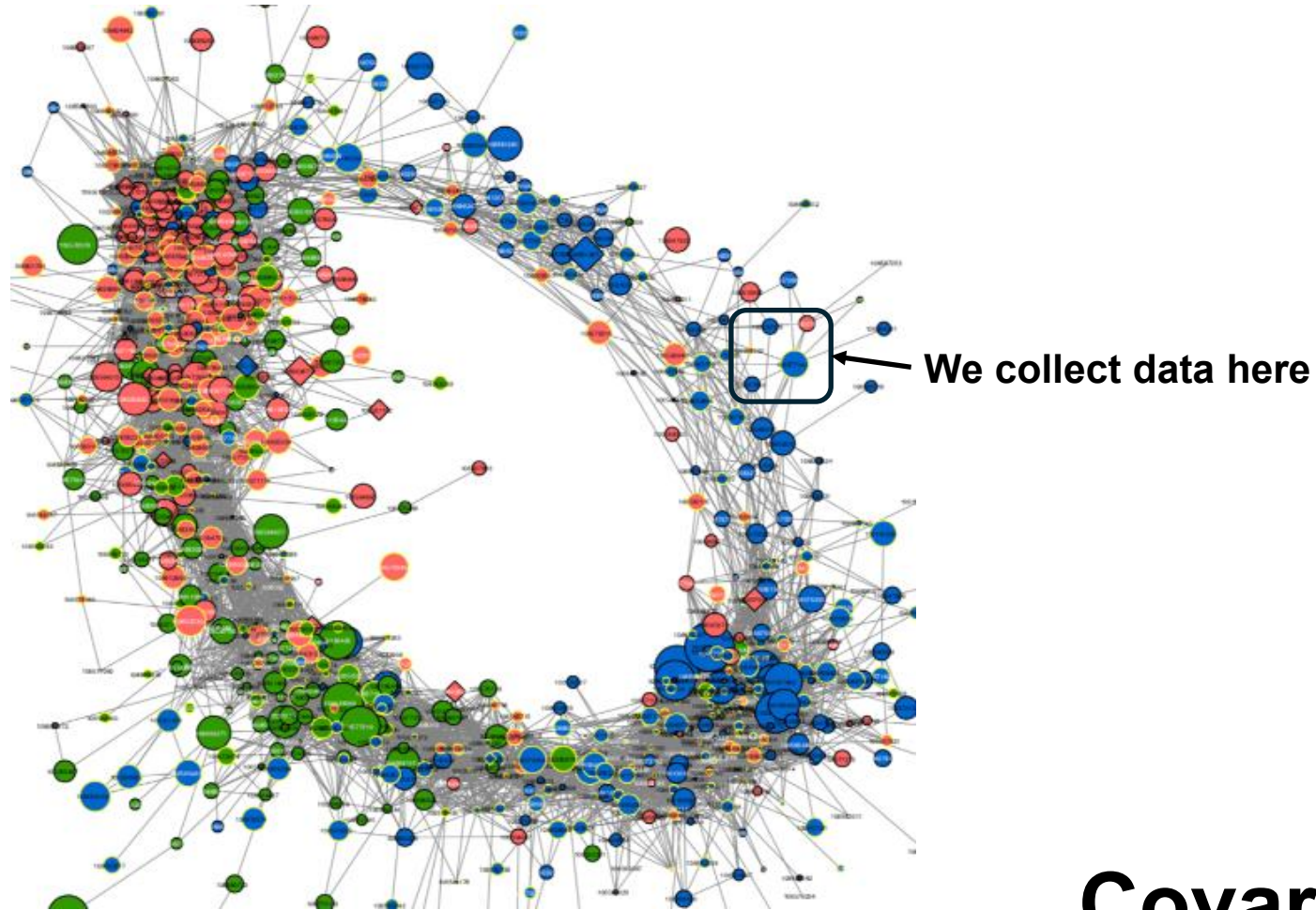
Figure 1.1
Well prepared cat.

GLMs focus on a single response as a function of covariates



Indirect effects: when an independent variable influences a dependent variable *via* one (or more) mediators

All models are wrong, but some are useful – GC Box

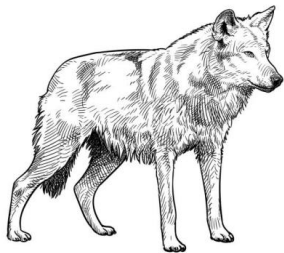
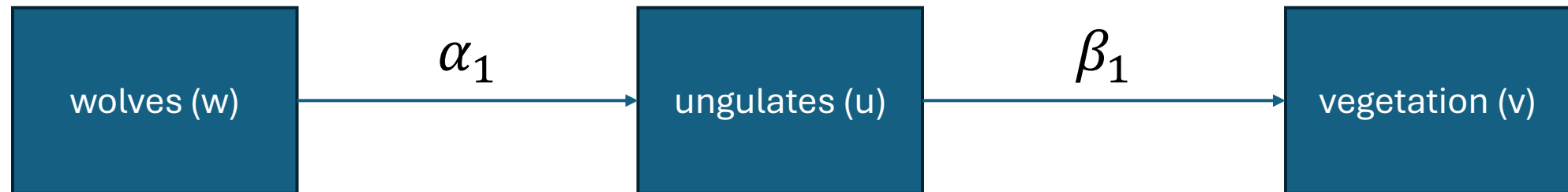
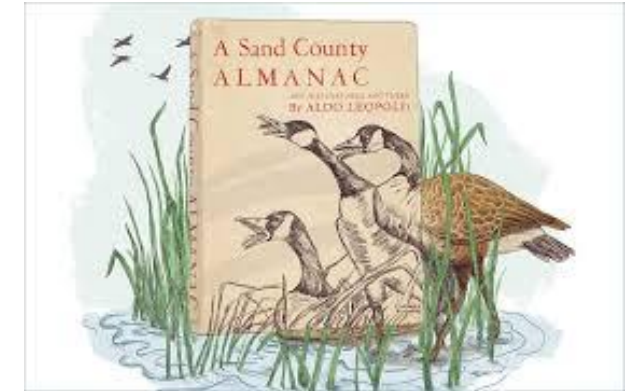


No models are right, and most are useless – TW Arnold

A simple example

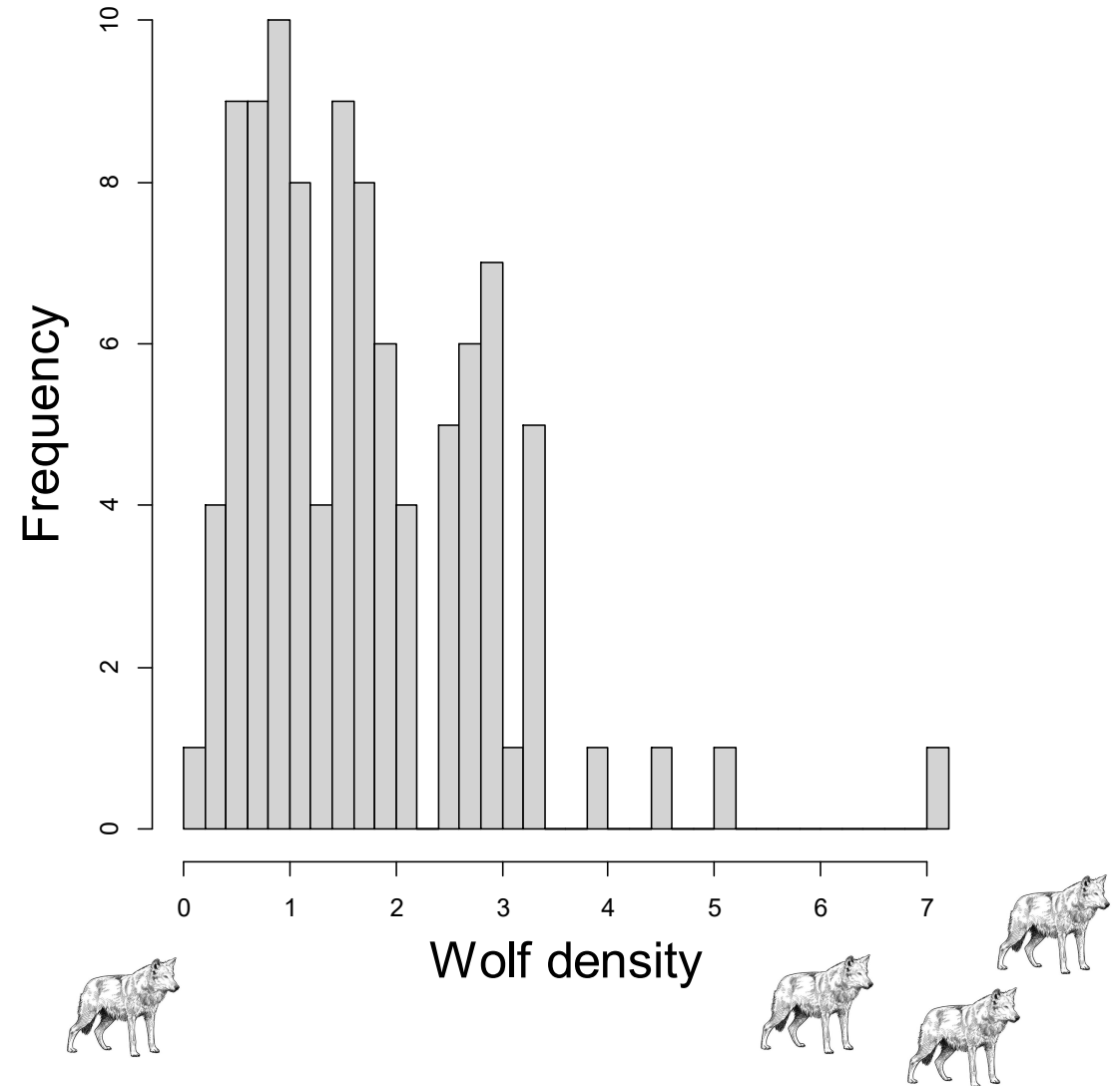
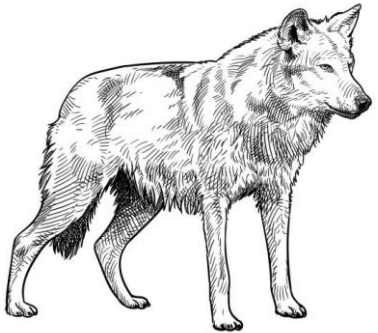
Today's dataset

Available browse ($\mathbf{v}$; vegetation) as a function of ungulate ($\mathbf{u}$) and predator ($\mathbf{w}$; wolf) abundance.



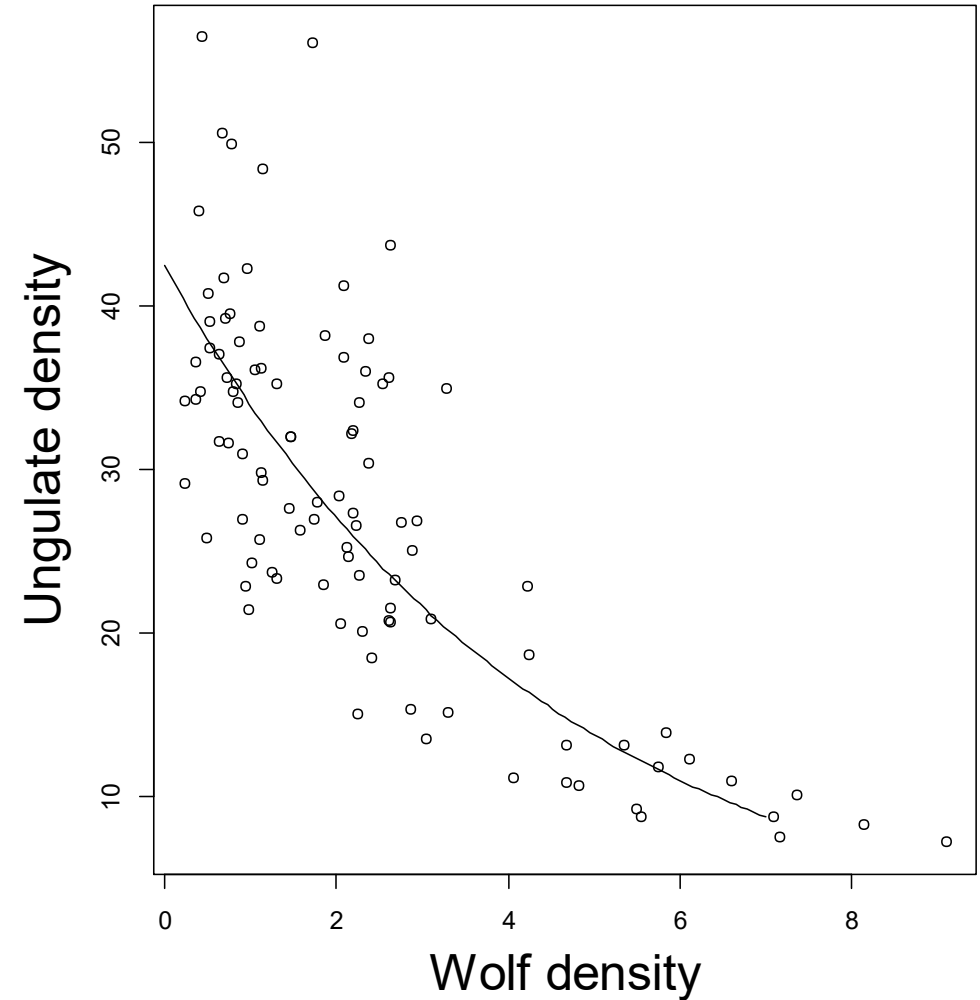
First, we'll simulate random variation in wolf density (w)

$$w_i \sim \text{lognormal}(0.35, 0.75)$$



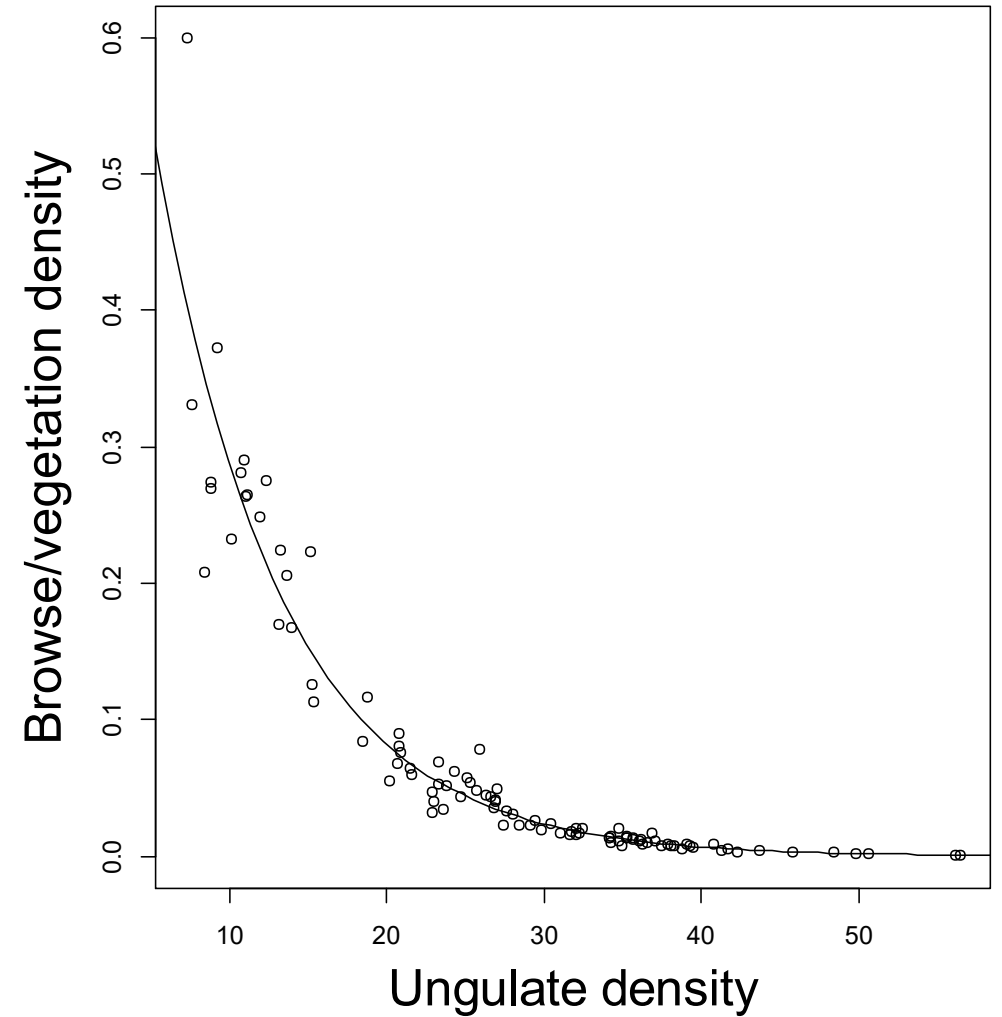
Second, we'll simulate ungulate density (u) as a function of w

$$u_i \sim \text{lognormal}(\alpha_0 + \alpha_1 w_i, \sigma_v^2)$$



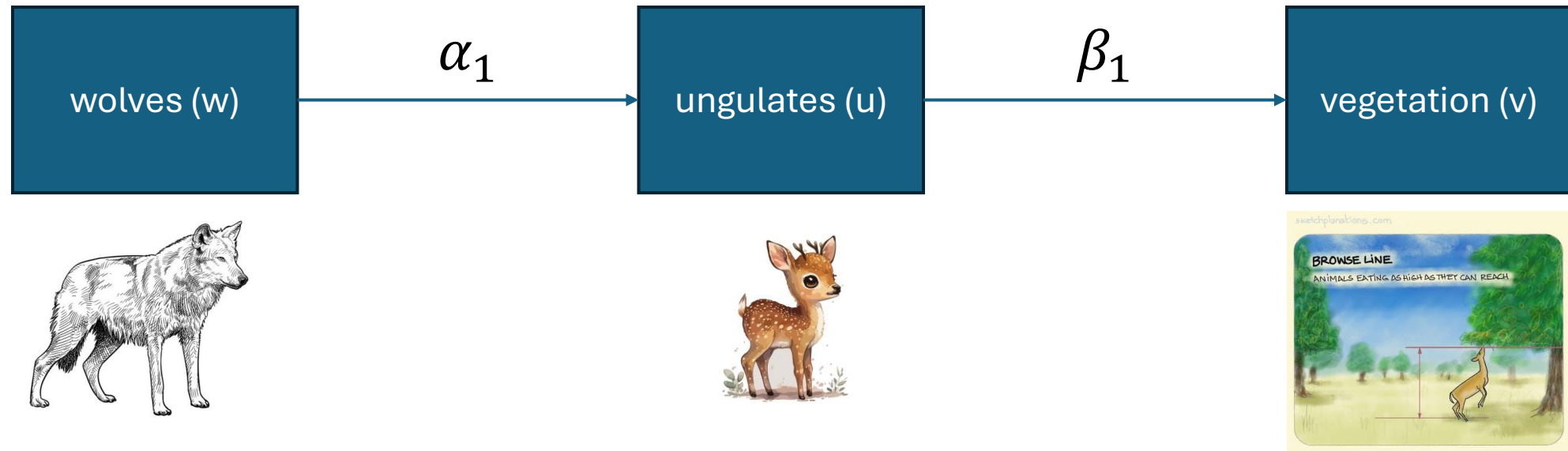
Third, we'll simulate browse density (v) as a function of u

$$v_i \sim \text{lognormal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



We'll use two (very similar) models to analyze the data

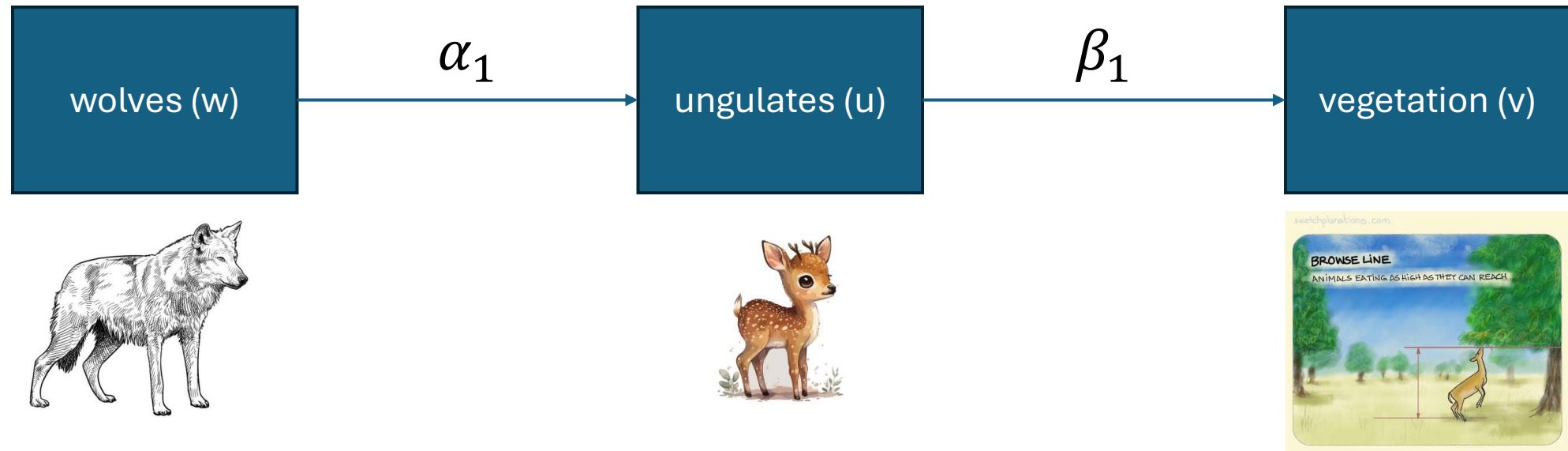
1. Identity link functions (i.e., normal distributions) in R
2. Log-normal distributions (to constrain to 'possible' values) in JAGS



First, we'll use 'identity' link functions in R [lm()]

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

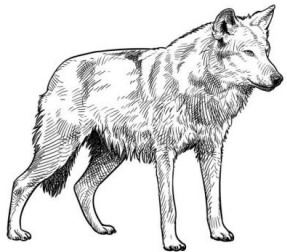
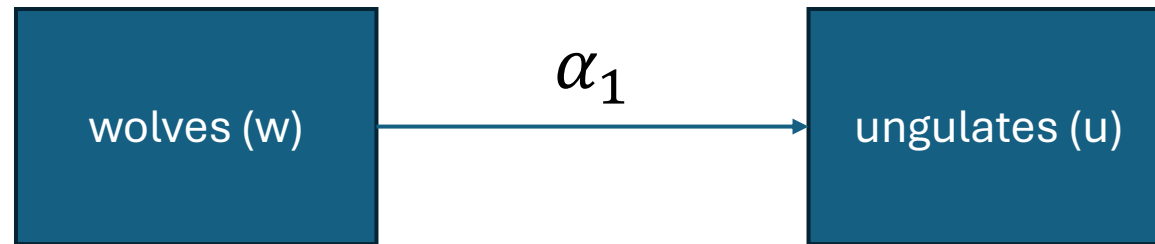
$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



Something to think about

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



$$\text{lm}(u \sim w)$$

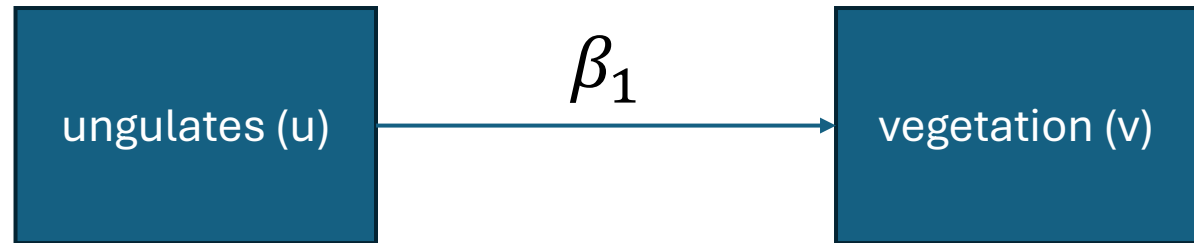
It's easy to calculate the effect of wolves on ungulate (α_1)!

Something to think about

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$

$$\text{lm}(v \sim u)$$

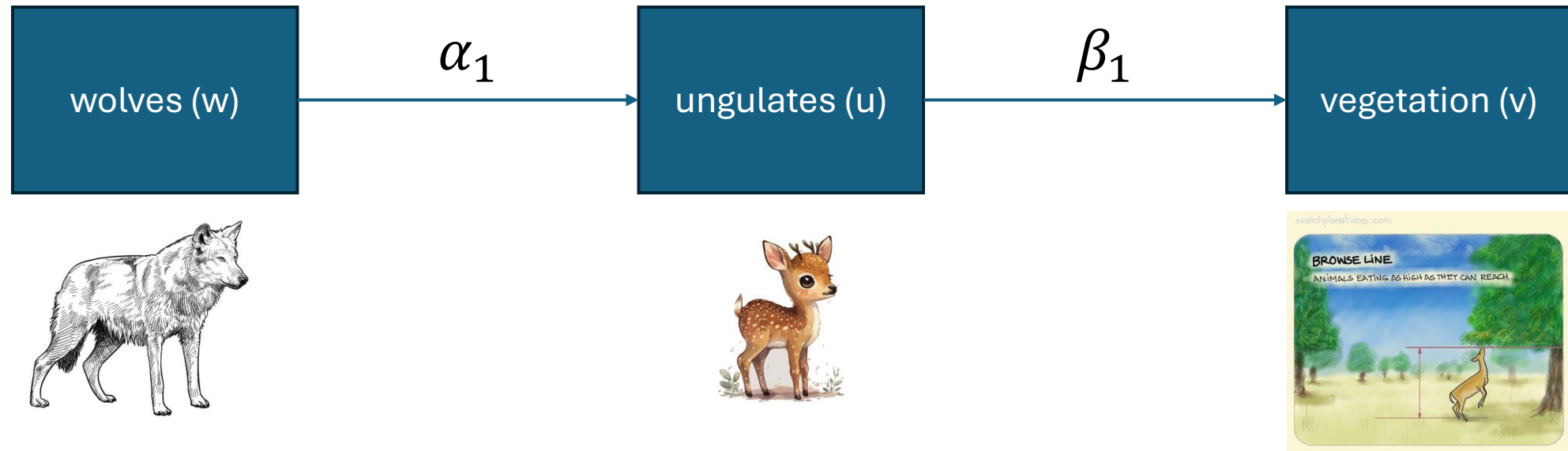


It's also easy to calculate the effect of ungulates on vegetation (β_1)

Something to think about (we'll discuss this shortly!)

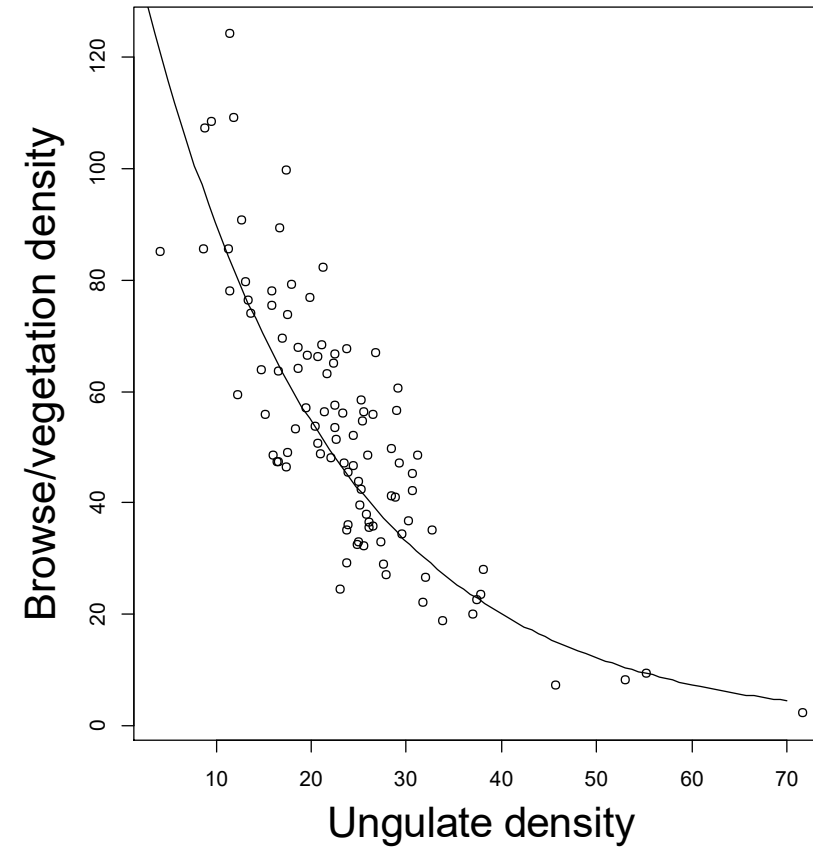
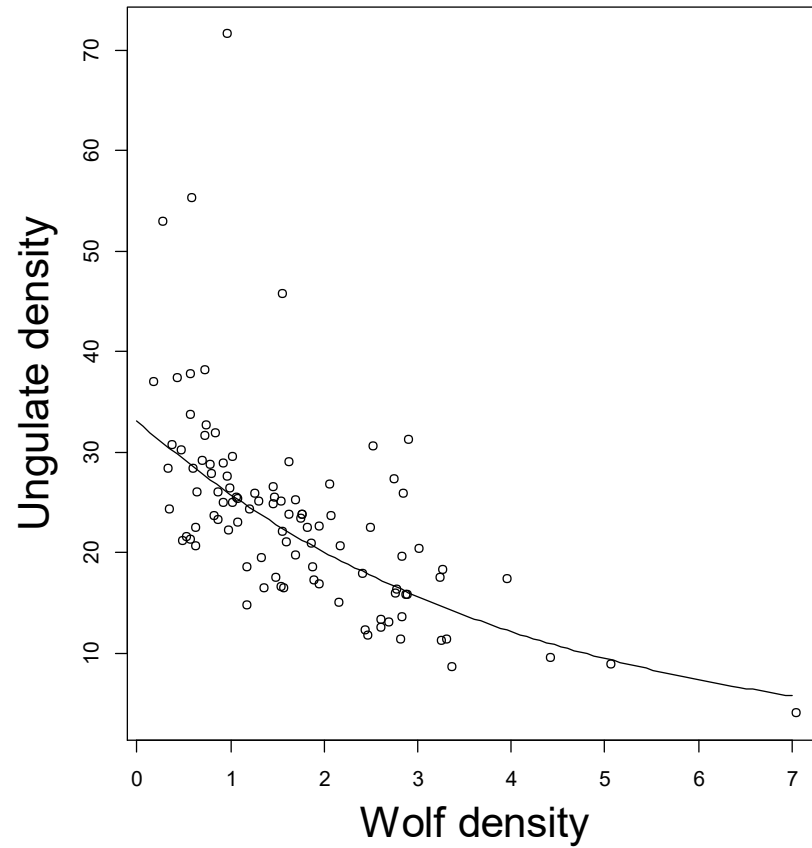
$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



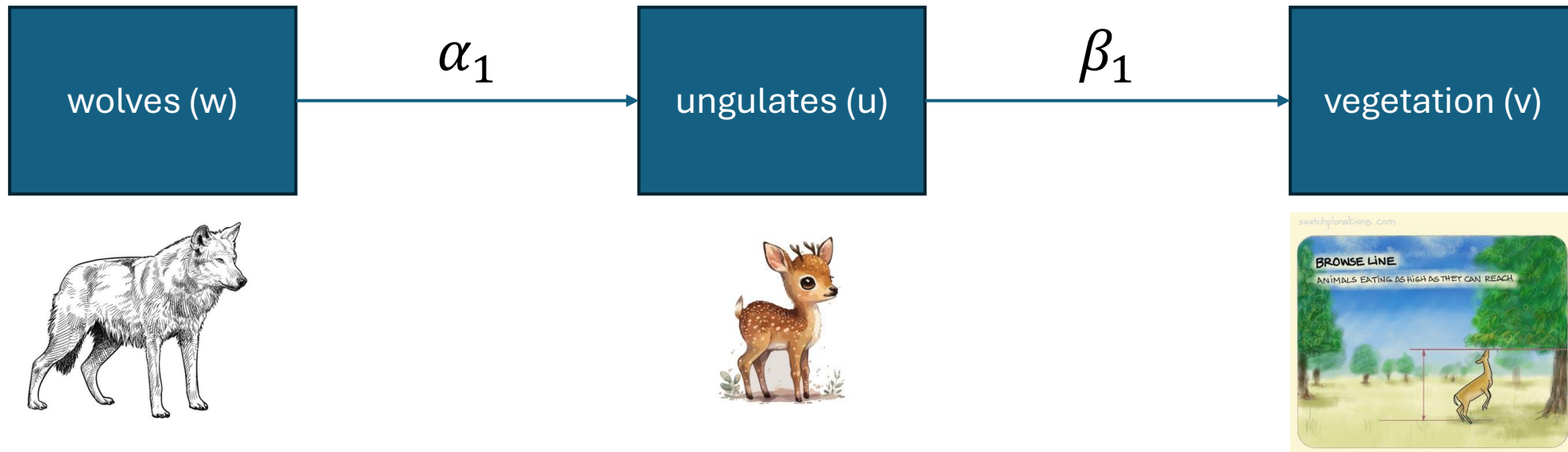
How would we calculate the effect of wolves on browse?!

How would we calculate the effect of wolves on browse?!



Let's imagine there is a hypothetical 101st study site with a current density of 2 wolves per 'unit'

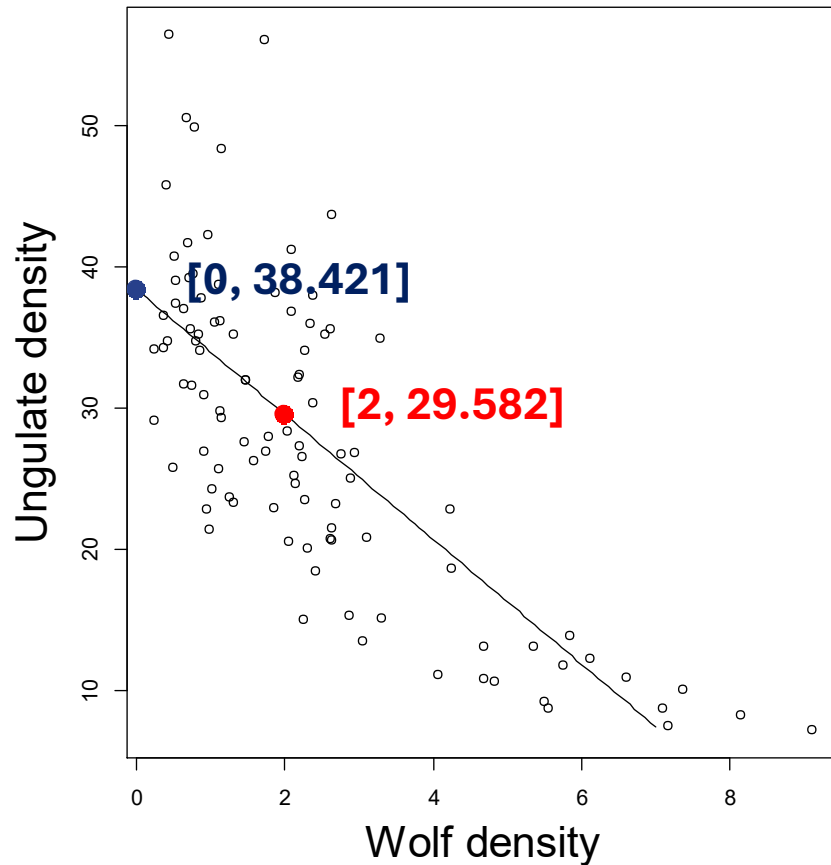
1. What is the current ML estimate of ungulates and browse?
2. What would happen to ungulate density if wolves were extirpated?
3. What would happen to browse if wolves were extirpated?



Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



We would gain about 8.83 ungulates, going from 29.582 to 38.421.

$$\alpha_0 = 38.4211$$

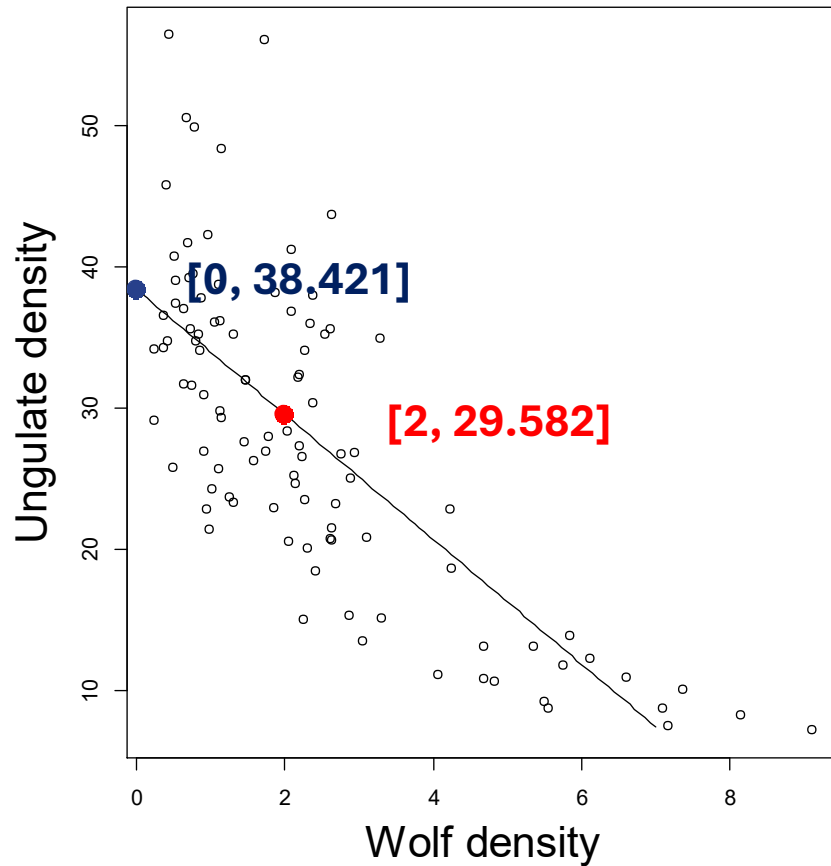
$$\alpha_1 = -4.4194$$

What would happen if we went from 2 wolves to 0?

Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



We would gain about 8.83 ungulates, going from 29.582 to 38.421.

$$\alpha_0 = 38.4211$$

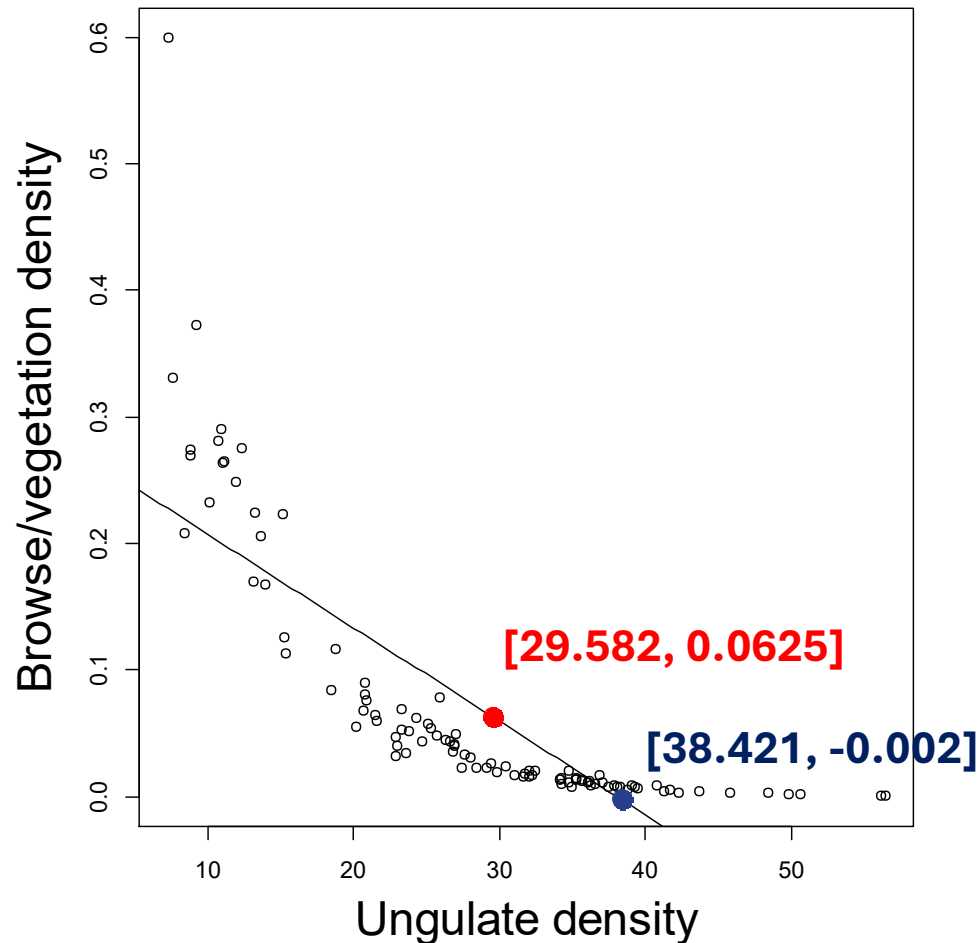
$$\alpha_1 = -4.4194$$

How would that 8.83 gain in ungulates affect browse?

Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



If we gain about 8.83 ungulates, we lose 'browse'.

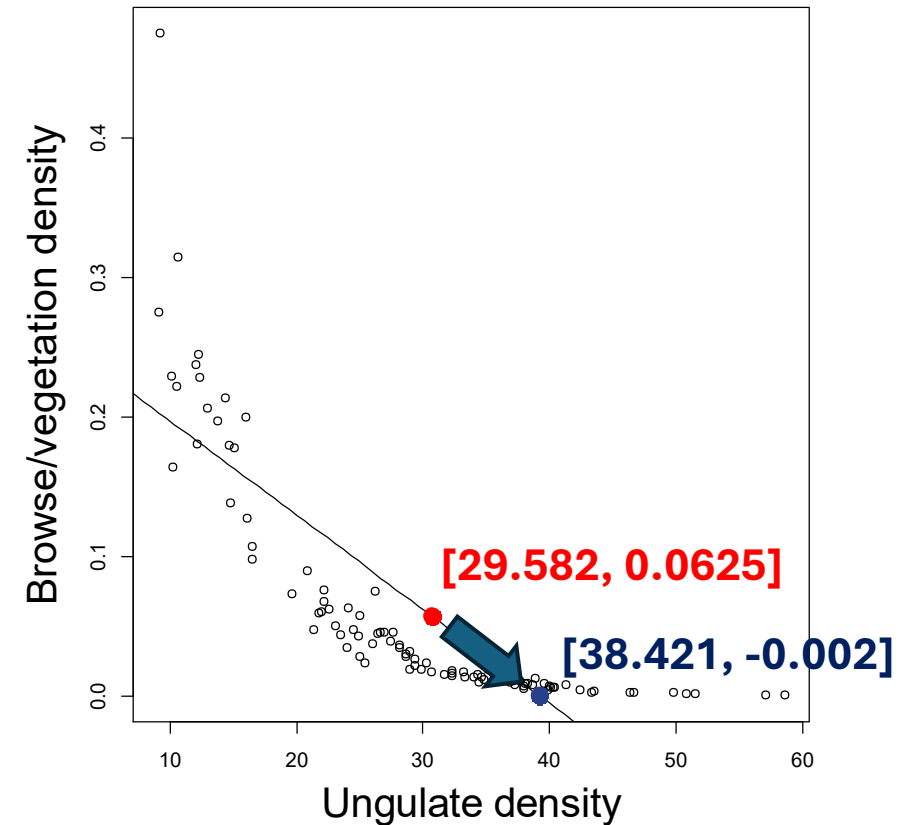
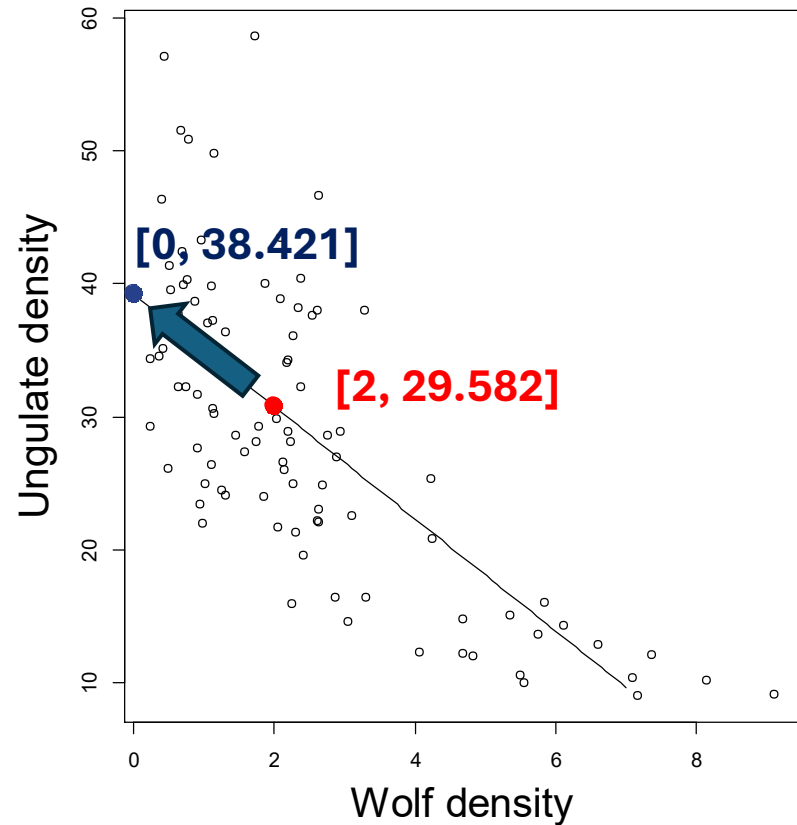
$$\beta_0 = 0.2809$$

$$\beta_1 = -0.00738$$

Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$

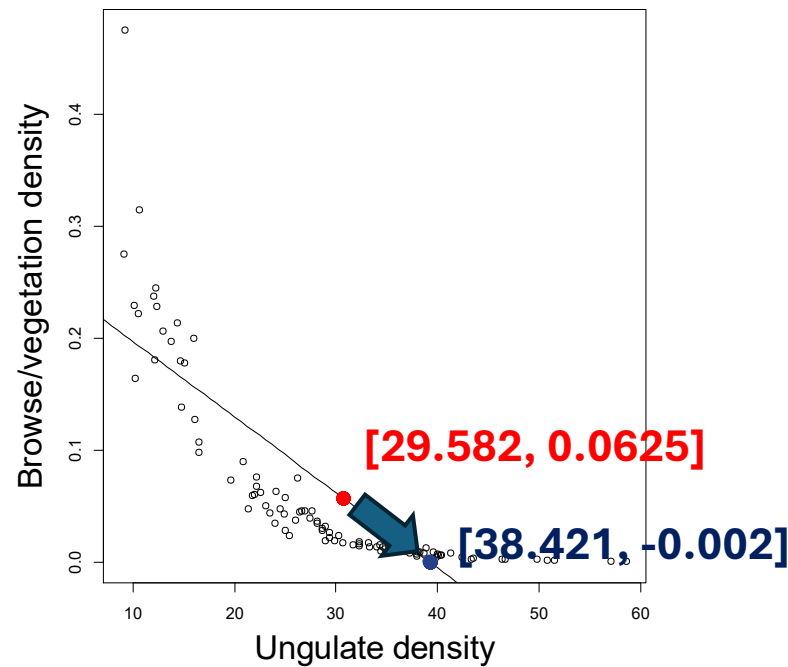
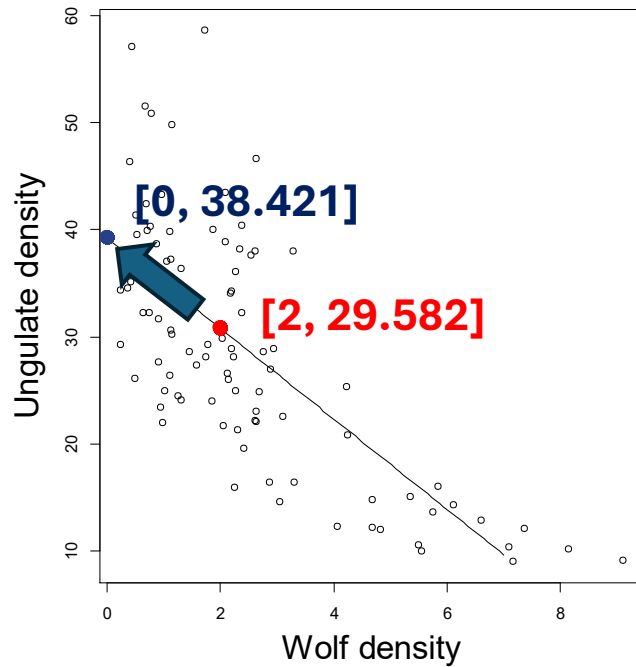


Losing wolves -> more deer; more deer -> less browse

Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



$$\beta_1 = -0.00738$$

$$\alpha_1 = -4.4194$$

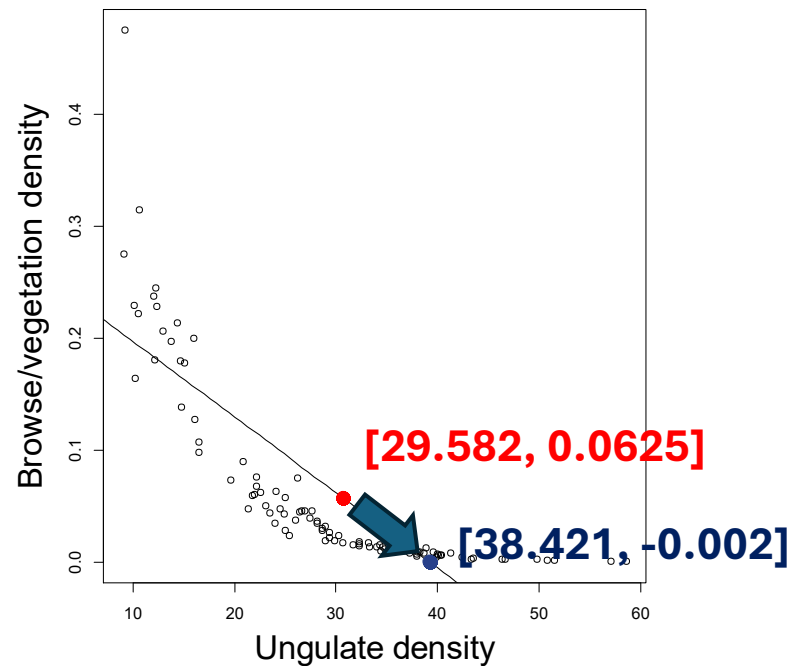
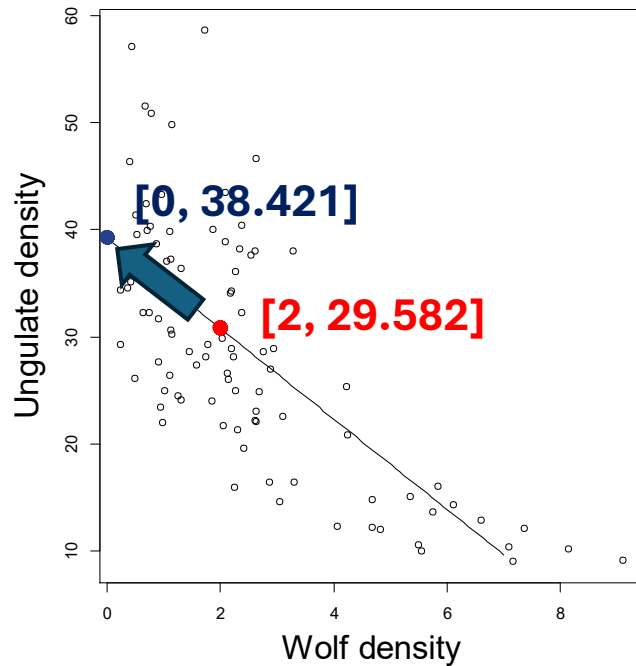
$$-0.06524 = -0.002 - 0.0625$$

Change in browse

Let's visualize an indirect effect!

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



$$\beta_1 = -0.00738$$

$$\alpha_1 = -4.4194$$

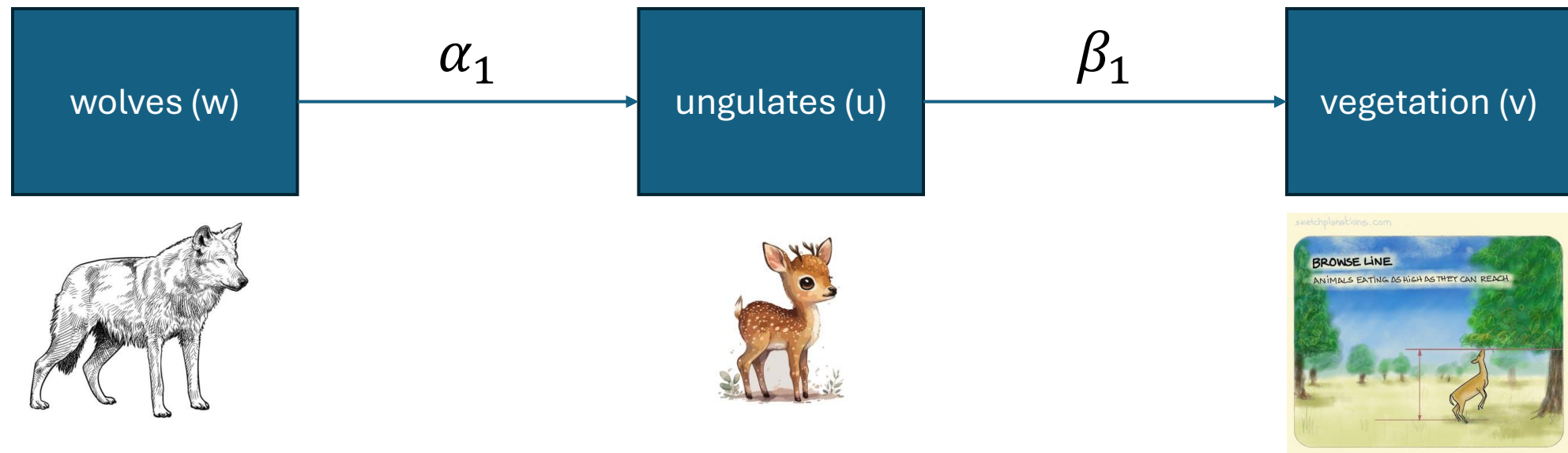
$$-0.06524 = \boxed{-0.002 - 0.0625} = \alpha_1(\beta_1)(-2)$$

Change in browse

Take-home 1: indirect effects aren't as 'horrifying' as they seem

$$u_i \sim \text{normal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{normal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



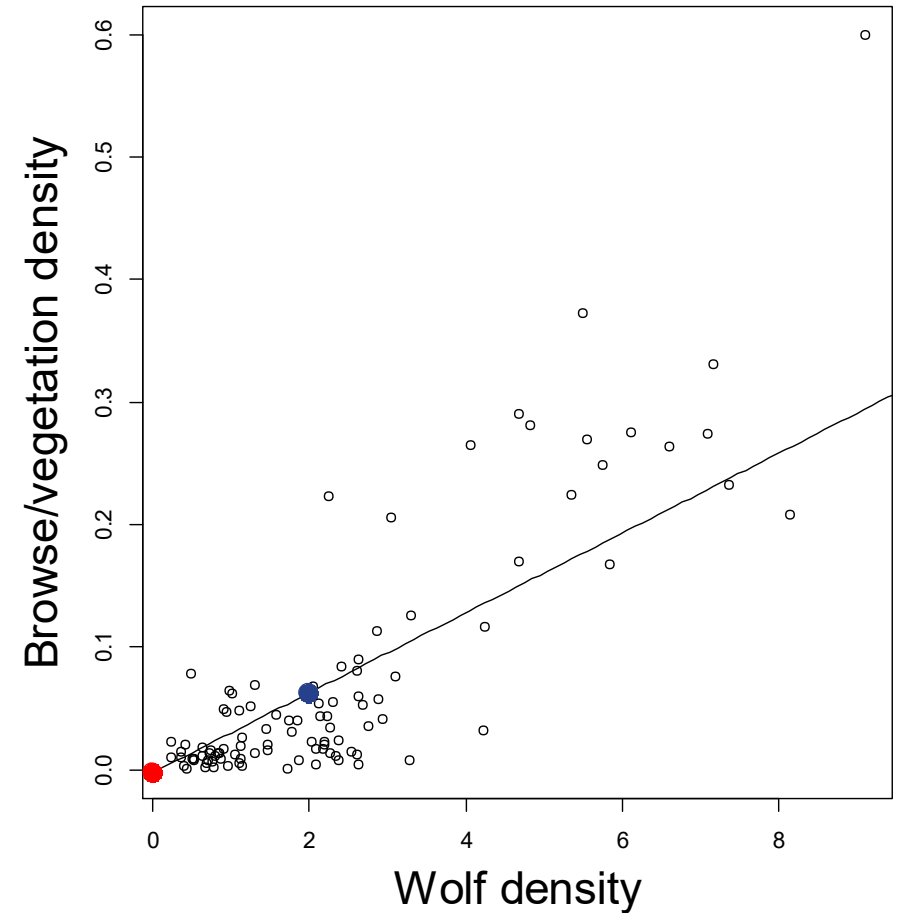
i.e., if identity link, then effect of wolves on veg = $\alpha_1 \beta_1$

$$E(v_i|w_i) = [\beta_0 + \beta_1(\alpha_0)] + \alpha_1\beta_1w_i$$

slope = $\alpha_1\beta_1$

intercept =

what value of veg would be
given number of deer if
there were no wolves

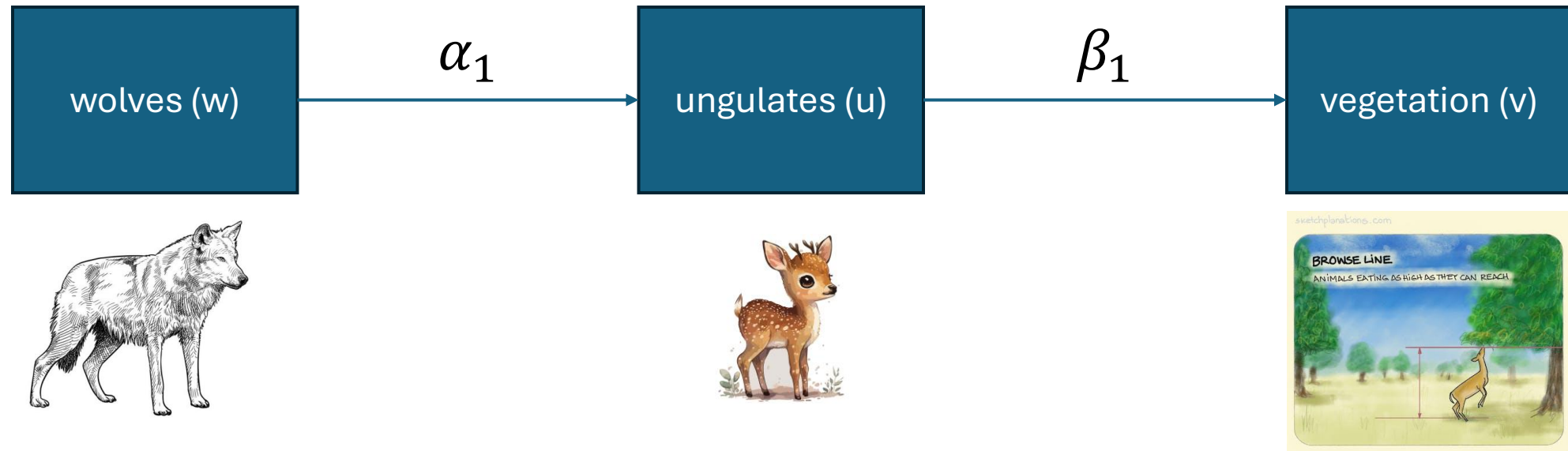


Let's go to the code and prove it!

Second, we'll use the data-generating model to analyze the data

$$u_i \sim \text{lognormal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

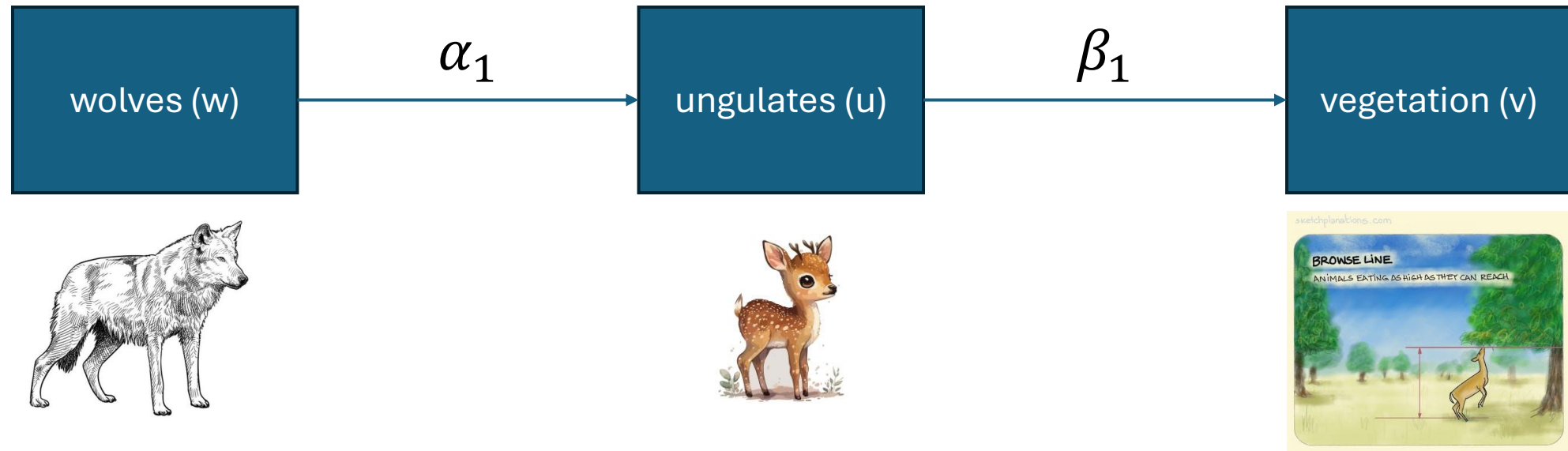
$$v_i \sim \text{lognormal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



Take-home: this takes some careful thinking!

$$u_i \sim \text{lognormal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{lognormal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



i.e., what link functions are we using, and how do we incorporate those?

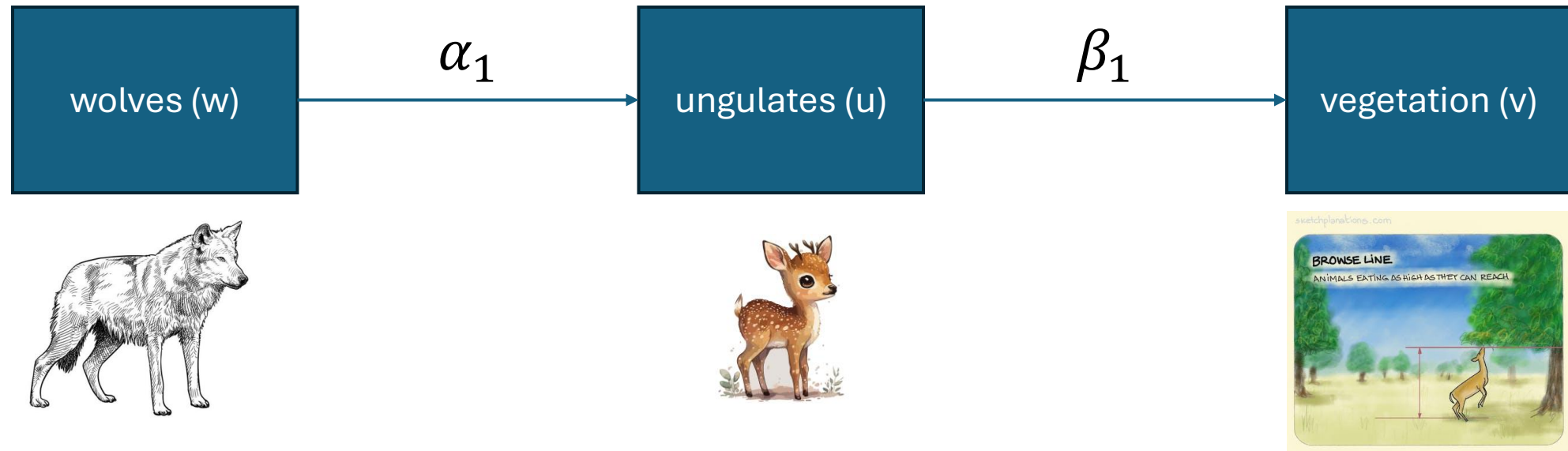
We may be transforming the y-axis (log- & logit-links)

i.e., what link functions are we using, and how do we incorporate those?

Take-home: this takes some careful thinking!

$$u_i \sim \text{lognormal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{lognormal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



i.e., are these z-standardized covariates? How do we back-transform axes?

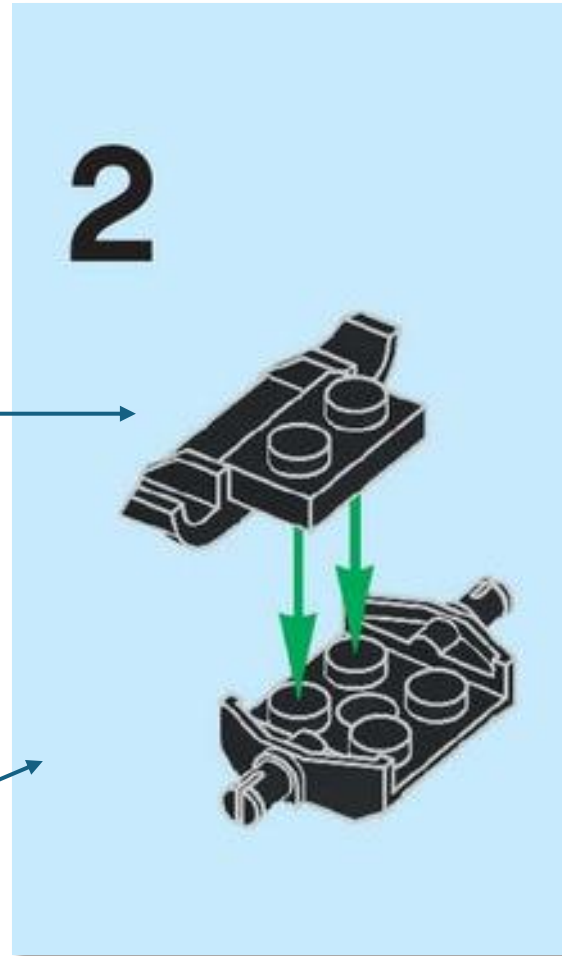
We may be transforming the x-axis (z-standardizing)

i.e., how are we scaling our covariates?

We have to keep track of a bunch of stuff

This piece is z-standardized

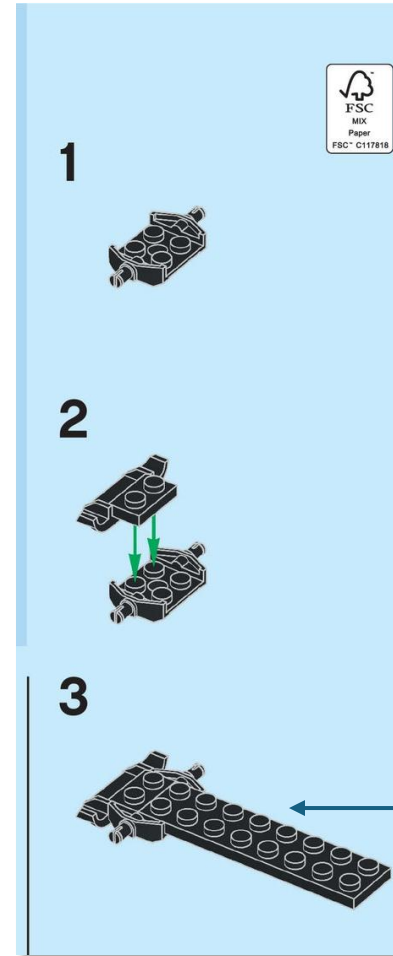
This piece is on the log-link



1

2

3

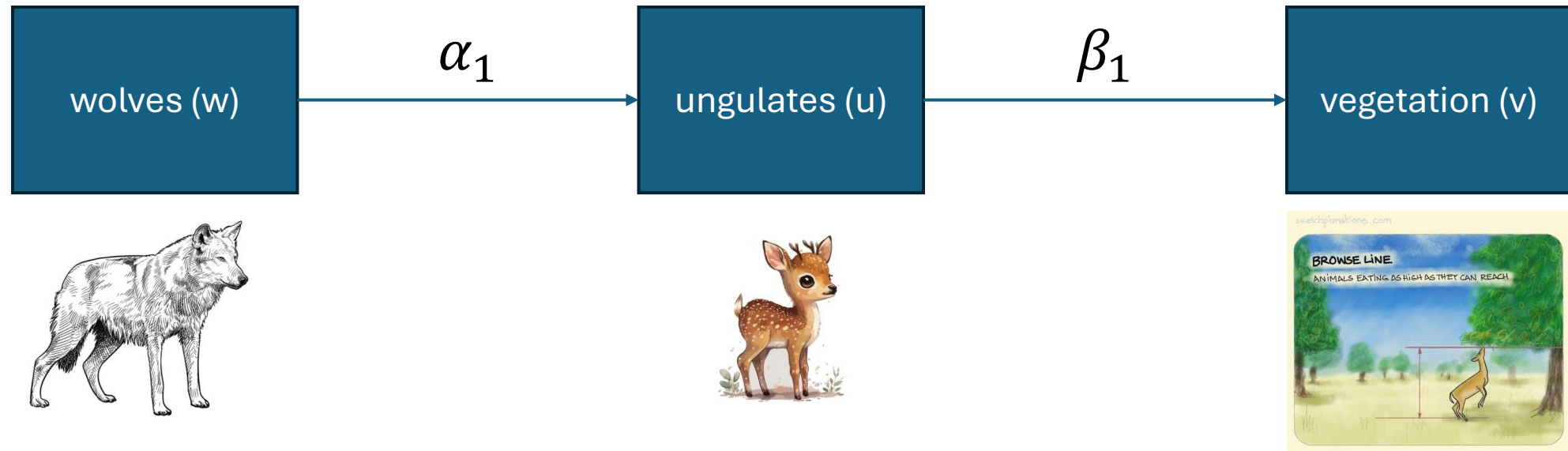


This piece is on the log-link

Take-home

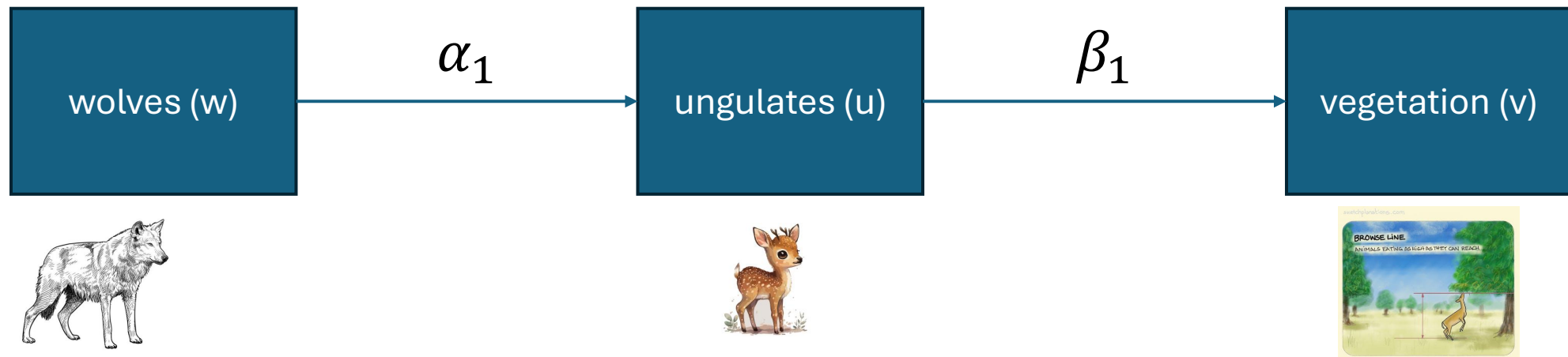
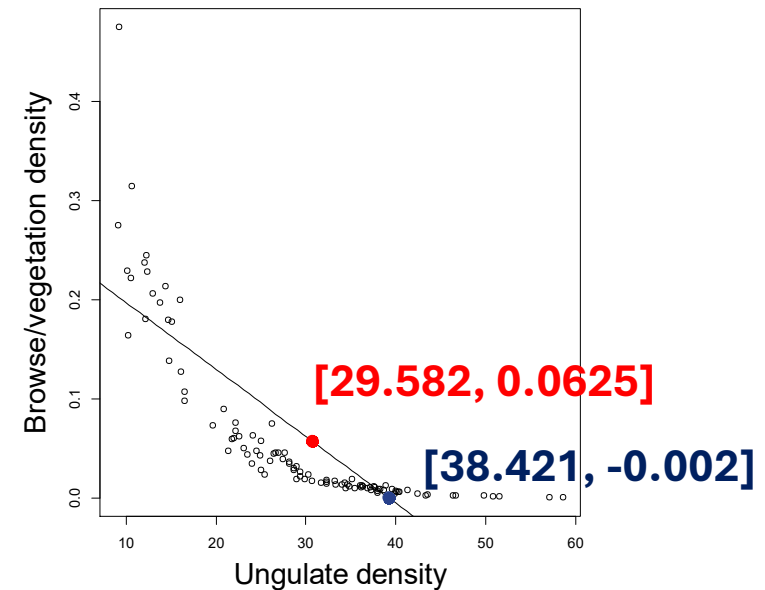
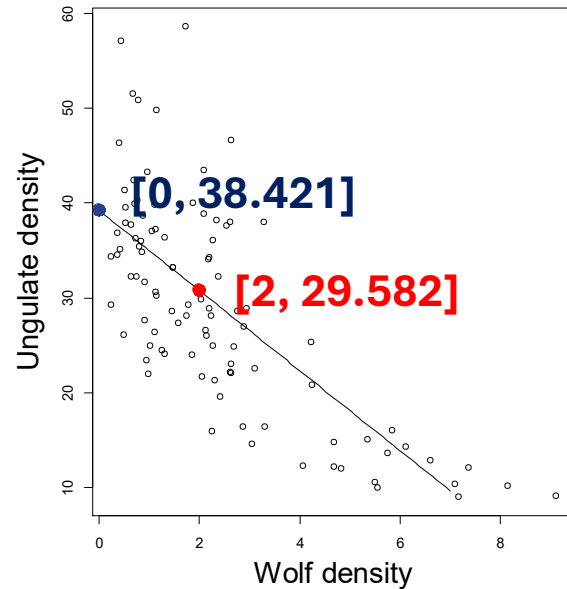
$$u_i \sim \text{lognormal}(\alpha_0 + \alpha_1 w_i, \sigma_u^2)$$

$$v_i \sim \text{lognormal}(\beta_0 + \beta_1 u_i, \sigma_v^2)$$



JAGS/Bayes will take some time, but it comes with advantages too!

Take-home: the math is tedious and requires careful thought, but the concept is simple and intuitive!



Take-home: on Thursday, we'll review and expand!

